



THE LEAN LAB *PLAYBOOK*

Cost Control & Efficiency Strategies
for the Modern Molecular Lab

The Lean Lab Playbook is a practical operational guide for lab managers, directors, and senior scientists. Nothing herein constitutes billing, legal, or compliance advice. Consult your billing team and legal counsel regarding reimbursement policies and payer contracts specific to your lab. BioPathogenix Products are RUO for Research Use Only. Not for use in diagnostic procedures.

THE LEAN LAB PLAYBOOK

Cost Control & Efficiency Strategies for the Modern Molecular Lab

The economics of pathogen detection and molecular diagnostics have changed. Reimbursement rates are under sustained pressure, coverage policies are tightening, and lab directors are being asked to do more with less. The playbook that worked five years ago — run broad panels, bill aggressively, maximize volume — is giving way to a new operating model built on precision, efficiency, and smart resource management.

This guide is designed to help your lab navigate that shift. Inside, you'll find a clear-eyed look at the current reimbursement environment, practical strategies for reducing waste and optimizing workflow, lab technique reminders that directly impact reagent costs, and a look at how modern reagent formats can be structured to support the way labs actually run today.

This is not a compliance guide, and nothing here constitutes billing or legal advice. It is a practical operational resource — built for lab managers, directors, and senior scientists who are ready to treat efficiency as a competitive advantage.

SECTION 1

Understanding the Current Reimbursement Environment

What's Changing — and Why It Matters

Reimbursement rates for molecular diagnostics and PCR-based testing have been declining steadily. This is driven by a combination of federal policy updates, payer-level coverage restrictions, and benchmarking practices that compress what labs can expect to collect per test.

Key Drivers of Reimbursement Pressure

Several converging forces are reshaping the reimbursement landscape:

- **CLFS & PAMA Benchmarking:** The Clinical Laboratory Fee Schedule (CLFS) sets Medicare payment rates for lab tests. Under the Protecting Access to Medicare Act (PAMA), these rates are periodically recalibrated based on private payer data — and in recent cycles, the recalibrations have trended downward for many molecular and PCR-based assays.
- **Coverage Narrowing:** Payers are becoming more specific about which targets qualify for reimbursement. Broad, reflexive panels that were once reimbursed in full are now being scrutinized, with payers approving only a subset of targets per run.
- **Well Limits Per Run:** Some payers now specify the number of billable wells per plate run, limiting the revenue per plate regardless of how many wells are physically run.
- **Rate Compression:** Commercial payers increasingly benchmark off lower-cost offerings or Medicare rates. Labs that relied on stronger commercial rates are seeing those advantages erode.

The Net Effect on Labs

Same sample volume. Same operational costs. Less revenue per test.

Labs that haven't adjusted their workflow and reagent strategies to match this new reality are effectively absorbing the rate cuts without offsetting them — which is why efficiency is no longer just an operational goal. It's a financial imperative.

How Labs Are Responding

The most forward-thinking labs are responding to this pressure in two main ways:

Plate & Panel Optimization

- Shifting to targeted panel configurations
- Testing for the most clinically relevant, commonly reimbursed targets

Operational Efficiency Improvements

- Conducting internal waste audits — where are reagents being discarded?

• Reducing panel breadth to avoid over-running • Aligning plate formats to actual sample volume and payer coverage • Running partial plates rather than full fixed formats to reduce waste	• Standardizing pipetting and aliquoting protocols • Reviewing calibration schedules and equipment maintenance logs • Renegotiating kit and reagent contracts based on actual usage patterns • Evaluating flexible reagent formats to reduce locked-in capacity waste
--	---



Note: Rate changes vary by payer, test type, and geography. Always consult your billing team and current payer contracts for figures specific to your lab.

SECTION 2

Where Labs Lose Money — The Four Waste Categories

Before you can cut costs, you have to know where they're leaking. In most molecular labs, waste concentrates in four areas. Some are obvious. Others hide in plain sight.

A. Fixed-Format Testing Waste

Pre-filled, fixed-format plates lock your capacity before you know what your sample flow looks like. When sample volumes fluctuate — as they always do — labs end up running plates with underutilized wells. Every empty or unused well represents reagent that was prepared, loaded, and discarded without generating any result or revenue.

- Pre-filled plates commit reagent to every position regardless of sample availability
- Underutilized wells generate no billable result — only cost
- Format mismatch between plate capacity and actual sample flow is one of the most common and underrecognized sources of reagent waste
- Labs running 24 or 48 wells on a 96-well plate are, in effect, paying for wells they're not using

Real-World Impact

A lab running 60-well plates on a 96-well format three times per week is effectively discarding 36 wells of reagent per run — roughly 37% waste before a single pipette error occurs.

At current reagent costs, that adds up quickly. And with reimbursements declining, the math only gets worse.

B. Over-Preparation and Over-Aliquoting

Preparing more reagent than you need seems like a safety measure. In practice, it's one of the most common sources of daily reagent loss. Over-aliquoting and over-preparation are baked into many lab SOPs that were written when reagent budgets were more forgiving. They need to be revisited.

- Running full panels when only a partial panel is clinically indicated generates results that can't be billed, using reagent that can't be recovered
- Excess reagent prepared for a run cannot be returned to stock — it is discarded
- Multi-freeze-thaw limitations on many wet reagents force labs to either use or lose prepared aliquots
- Bulk mixing in anticipation of volume that doesn't materialize is a silent budget drain that rarely shows up on waste logs

C. Workflow Drift

Workflow drift is what happens when the gap between your written SOP and actual bench practice widens — slowly, invisibly, and expensively. It's not negligence. It's the natural result of staff turnover, informal training, pressure to move faster, and the assumption that experienced technicians don't need refreshers.

Calibration Inconsistencies

Pipettes that are even slightly out of calibration deliver inaccurate volumes with every use. This affects both the reliability of results and the efficiency of reagent use — under-delivering wastes reagent through failed runs; over-delivering depletes stock faster than your procurement model anticipates.

- Pipettes should be calibrated at regular intervals — typically every 3 to 6 months depending on usage volume
- Post-repair or post-drop calibration is critical and often skipped
- Calibration records should be logged and auditable
- Single-channel and multi-channel pipettes have different calibration drift profiles and should be assessed separately

Pipetting Variability

Technique matters more than most labs account for in their cost models. Variability in pipetting angle, aspiration speed, tip submersion depth, and air gap practices all affect delivered volume — and therefore reagent use and result reliability.

- Reverse pipetting technique is recommended for viscous reagents and master mix preparation
- Pre-wetting tips before aspirating improves volume accuracy, especially at low volumes
- Consistent aspiration speed reduces bubble formation and volume error
- Touch-off against the tube wall after dispensing ensures full volume delivery
- Tip changes between samples are non-negotiable for contamination prevention — but tip-change discipline also affects run efficiency and should be assessed

Training Gaps

Labs often assume that once staff are trained, they stay trained. The reality is that technique degrades, shortcuts are adopted, and new hires inherit the informal habits of whoever trained them — not the habits in the SOP.

- Annual or semi-annual technique refreshers pay for themselves in reagent savings
- Competency assessments should include direct observation of pipetting technique, not just written tests
- Cross-training between team members surfaces technique inconsistencies that internal siloing hides

D. Inventory Rigidity

Ordering in bulk formats that don't match your actual usage patterns creates a persistent mismatch between what you have on the shelf and what you need on the bench. This leads to two problems: short shelf-life pressure that forces premature use or disposal, and over-ordering that ties up budget in inventory that degrades before it's used.

- Wet reagent kits with shorter shelf lives create pressure to run tests before they're needed — inverting the clinical workflow
- Bulk formats ordered to hit pricing thresholds may cost less per unit but more in total if significant quantities expire
- Lack of flexibility in kit sizing forces labs to choose between over-ordering and under-ordering — neither is efficient
- Domestic sourcing and shorter lead times allow for more responsive, lower-inventory purchasing models

SECTION 3

Efficiency Strategies That Actually Move the Needle

This section moves from diagnosis to prescription. These are the specific strategies that labs in constrained reimbursement environments are using to reduce per-run costs and improve operational predictability.

Strategy 1: Align Your Plate Configuration to Your Sample Flow

The most impactful operational shift a lab can make is aligning its plate format and panel configuration to its actual, observed sample flow — not its peak capacity, and not its legacy format.

- Audit your last 90 days of runs: What was the average fill rate per plate? What targets were run most frequently? Which wells were most often unused?
- Identify your highest-volume, most consistently reimbursed targets and build your primary panel around those
- Use the panel breadth flexibility your reimbursement environment actually allows — not the broadest panel your platform technically supports
- Consider whether your current plate format (96-well, 48-well, 24-well) matches your median sample volume, not your maximum

Practical Tip: The 80/20 Panel Review

In most labs, 80% of billable volume comes from 20% of the panel targets. Identify your core 20% and build your default configuration around it.

This doesn't mean abandoning low-frequency targets — it means running them intentionally, on plates sized to match their volume, rather than including them in every run by default.

Strategy 2: Move Toward Modular Reagent Formats

One of the most structurally effective ways to reduce fixed-format waste is to use reagent formats that allow you to run only what you need — without committing the entire plate.

Lyophilized reagents in individually detachable tube formats represent a meaningful operational shift from traditional wet plate configurations. Instead of opening an entire plate and committing all wells to a run, labs can detach only the tubes they need for the samples in front of them.

- Unused tubes stay sealed and stable — no reagent is wasted on wells that weren't run
- Partial runs become standard practice rather than exceptions requiring workarounds
- Extended shelf life (BPX lyoplates carry an 18-month shelf life) reduces the short shelf-life pressure that forces premature runs

- Tolerance for up to 5 freeze-thaw cycles on lyophilized formats provides operational flexibility that wet reagents cannot match
- No order minimums allow labs to order closer to actual usage rather than bulk-ordering to hit pricing thresholds

Why This Matters in the Current Environment

When payers limit the number of reimbursable wells per run, running a full 96-well plate is not just inefficient — it's generating costs you can't recover.

Modular formats let your reagent use track your reimbursable output. That alignment is the foundation of sustainable cost control.

Strategy 3: Right-Size Your Panel Breadth

The shift in payer behavior — covering fewer targets per run, scrutinizing broad reflexive panels — makes panel breadth a cost control lever, not just a clinical decision.

- Review your panel against your current payer contracts: Which targets are consistently covered? Which are being denied or downgraded?
- Work with your medical director and billing team to identify the clinically indicated, reimbursement-supported panel configuration for your primary patient population
- Consider tiered panel configurations: a core panel for routine runs, and supplemental configurations for specific clinical indications
- Document clinical indication for expanded panels — this is both good practice and a necessary defense if coverage decisions are audited

This is not about reducing clinical thoroughness. It's about ensuring that what you run is supported by both clinical indication and payer coverage — so you're not absorbing the cost of tests that generate neither billable results nor clinical action.

Strategy 4: Implement a Reagent Waste Audit

You can't manage what you don't measure. A structured reagent waste audit — even a simple one — surfaces the specific points in your workflow where reagent is being prepared and discarded.

What to Measure

- Volume of reagent prepared vs. volume loaded per run
- Volume loaded vs. volume consumed (accounting for dead volume in reservoirs and tips)
- Number of unused wells per plate per run, averaged over 90 days
- Reagents discarded due to expiration vs. reagents consumed in active use
- Frequency and volume of reagent discarded due to failed QC or invalid runs

What to Do with the Data

- Map waste to specific workflow steps — preparation, loading, storage, or run failure
- Prioritize the highest-volume waste sources for process review
- Set targets and track improvement over 30, 60, and 90 days
- Share the data with your reagent vendors — reagent usage patterns can inform kit sizing, format discussions, and procurement planning

SECTION 4

Lab Technique & Etiquette — The Fundamentals That Fund Themselves

Technique reminders can feel like basics — and they are. But the basics compound. A lab with consistently precise pipetting technique, properly calibrated equipment, and well-maintained SOPs runs fewer failed plates, wastes less reagent, and generates cleaner data. In a constrained reimbursement environment, those differences add up to real money.

Pipetting Best Practices

Volume Accuracy Fundamentals

- Always use calibrated pipettes. Check calibration dates before use and report any pipette that has been dropped or shows signs of mechanical wear.
- Hold the pipette vertically when aspirating. Tilting beyond 20° introduces volume error that compounds across every well.
- Aspirate smoothly and at a consistent speed. Rapid aspiration pulls air into the tip and reduces delivered volume.
- Touch the tip to the inner wall of the tube at a 10–45° angle when dispensing. This ensures full volume delivery without creating bubbles.
- After dispensing, do not blow out the last drop unless you are using forward pipetting technique with a calibrated blowout volume. With reverse pipetting, the residual volume is accounted for in the calibration.

Technique Selection by Application

Forward Pipetting • Standard aqueous solutions • Routine sample transfers • Volume ranges pipette is calibrated for • Most common technique — calibrations assume it	Reverse Pipetting • Viscous reagents (master mix, glycerol-containing solutions) • Volatile or foaming liquids • Very small volumes where bubble formation is a concern • Situations where tip coating causes volume loss
---	--

Pipette Calibration — A Cost Control Tool

Pipette calibration is not just a compliance checkbox. It is a direct input into reagent cost per run. A pipette delivering 5% below its set volume across a 96-well plate is generating 5% reagent waste on every run — while also producing results that may fail QC.

- Calibrate pipettes every 3 to 6 months under normal use conditions. High-volume labs or labs using aggressive reagents should calibrate more frequently.
- Always calibrate after a pipette is dropped, repaired, or returned from service.

- Use a calibrated analytical balance and distilled water for in-house verification. Weigh the volume delivered across 10 replicates and calculate CV.
- A CV above 1–2% for standard volumes indicates the pipette requires professional servicing.
- Keep a calibration log for every pipette in the lab. Auditable records are required for accreditation and are essential for troubleshooting systematic errors.
- Multi-channel pipettes should be verified across all channels — channel-to-channel variability is common and is the most frequent source of systematic error in plate-based assays.

Quick Calibration Check (In-House Verification)

1. Set pipette to target volume (e.g., 100 μ L)
2. Tare analytical balance
3. Pipette 10 replicates of distilled water onto tared balance
4. Record each reading in mg (1 mg $\approx$ 1 μ L for water at room temp)
5. Calculate mean and CV
6. Acceptable: Mean within $\pm 1\%$ of target; CV $< 1\%$ for most applications
7. Out of spec: Send for professional calibration — do not continue use

Equipment Maintenance That Affects Run Cost

Thermocycler & PCR System Performance

- Run temperature verification profiles at scheduled intervals — temperature uniformity across the block directly affects assay sensitivity and specificity
- Block non-uniformity causes well-to-well variability that can appear as failed samples, requiring re-runs that consume additional reagent
- Maintain ramp rate validation records. Degraded ramp rates extend cycle times without affecting results initially, but indicate block or Peltier degradation that will eventually cause failures
- Clean the block surface and lid between heavy-use cycles to prevent condensate contamination

Liquid Handling Systems (Where Applicable)

- Robotic liquid handlers require regular tip volume verification, equivalent to manual pipette calibration
- Dead volume in liquid handler reservoirs and tubing can be significant — account for it in reagent preparation volumes
- Regular maintenance cycles for wash steps and seal replacement prevent cross-contamination that causes run failures

Cold Storage and Reagent Integrity

- Verify freezer temperatures with a calibrated probe separate from the unit's built-in display. Freezer thermostats drift, and the unit's display does not always reflect actual sample compartment temperature.

- Avoid storing reagents in the door of a refrigerator or freezer — temperature fluctuates most at door locations
- Label all reagents with open date, expiration date, and number of freeze-thaw cycles used. For reagents with freeze-thaw limits, tracking begins at first thaw, not first use.
- Ensure freeze-thaw cycle limits are clearly posted next to the storage location — not just in the SOP

Aliquoting Best Practices

Aliquoting is one of the most impactful — and most inconsistently performed — steps in the workflow. Done well, it preserves reagent integrity and reduces freeze-thaw-related degradation. Done poorly, it wastes reagent, introduces contamination risk, and creates inconsistency across runs.

- Aliquot working volumes that match your actual run sizes — not the largest run you might ever do
- Label aliquots with date, reagent lot number, volume, and freeze-thaw count
- Avoid repeated freeze-thaw by using aliquot sizes that can be consumed in a single session
- Thaw reagents completely and mix by gentle inversion before use — do not vortex PCR reagents
- Work on ice when preparing aliquots of temperature-sensitive reagents to minimize degradation during handling
- After aliquoting, immediately return the master stock to storage — do not leave it on the bench

SECTION 5

The Importance of Selecting the Right Assay | Reagent Partner

Efficiency improvements don't require a total lab overhaul. The most effective approach is incremental and data-driven: identify your biggest waste sources, address them systematically, and track the results.

How Reagent Partners Can Support This Work

Your reagent vendors are a resource in this process — or they should be. The right partner can offer more than product. They can offer format flexibility, usage data, and supply chain responsiveness that supports a leaner operating model.

When evaluating your reagent relationships in the context of efficiency, consider:

- **Format flexibility:** Can your vendor supply reagents in formats that match your actual run sizes, rather than locking you into fixed-capacity configurations?
- **Shelf life:** Longer shelf-life reagents reduce the pressure to run tests before they're clinically indicated. Lyophilized formats typically offer significantly extended shelf life over wet reagents.
- **Freeze-thaw tolerance:** Reagents with higher freeze-thaw cycle tolerances give you more operational flexibility in how you manage stock.
- **Order flexibility:** No-minimum ordering allows you to procure closer to actual usage, reducing over-inventory and expiration loss.
- **Lead times:** Domestic production and distribution typically means faster replenishment, enabling lower on-hand inventory without stockout risk.
- **Custom configurations:** Labs with specific panel requirements benefit from vendors who can structure kit contents to match clinical workflow rather than forcing labs to adapt their workflow to a fixed kit.

A Note on BPX

BPX's reagent line is built with these efficiency pressures in mind. Our lyophilized plate format (96 and 384-well) with individually detachable tubes is designed to let labs run only what they need — eliminating the forced commitment to a full plate.

An 18-month shelf life, tolerance for up to 5 freeze-thaw cycles, no order minimums, domestic production, and custom kit capabilities make it possible to align your reagent strategy with the way your lab actually runs.

If you'd like to see how this maps to your specific workflow and panel configuration, we're glad to walk through it.

The Bottom Line

The labs that come through this reimbursement cycle strongest won't be the ones that cut the most — they'll be the ones that run the smartest. Precision over volume. Flexibility over fixed formats. Data over assumption.

This guide is a starting point. The data in your own run logs is the next step.

Questions? Ready to evaluate your current workflow?

Contact BioPathogenix to speak with a lab efficiency specialist.

*This document is for informational and operational purposes only. Nothing herein constitutes billing, legal, or compliance advice. Consult your billing team and legal counsel regarding reimbursement policies and payer contracts specific to your lab.
BioPathogenix Products are RUO for Research Use Only. Not for use in diagnostic procedures.*